

SAT 10
Math Module 1

Question # **ID**
SAT10 M 1.1 b8150b17

For a particular machine that produces beads, 29 out of every 100 beads it produces have a defect. A bead produced by the machine will be selected at random. What is the probability of selecting a bead that has a defect?

- A. $\frac{1}{2,900}$
- B. $\frac{1}{29}$
- C. $\frac{29}{100}$
- D. $\frac{29}{10}$

SAT10 M 1.2 c4d49134

$$s = 40 + 3t$$

The equation gives the speed s , in miles per hour, of a certain car t seconds after it began to accelerate. What is the speed, in miles per hour, of the car 5 seconds after it began to accelerate?

- A. 40
- B. 43
- C. 45
- D. 55

SAT10 M 1.3 a10ma103

If $4x = 3$, what is the value of $24x$?

☐ A $\frac{9}{2}$

☐ B 6

☐ C 18

☐ D 72

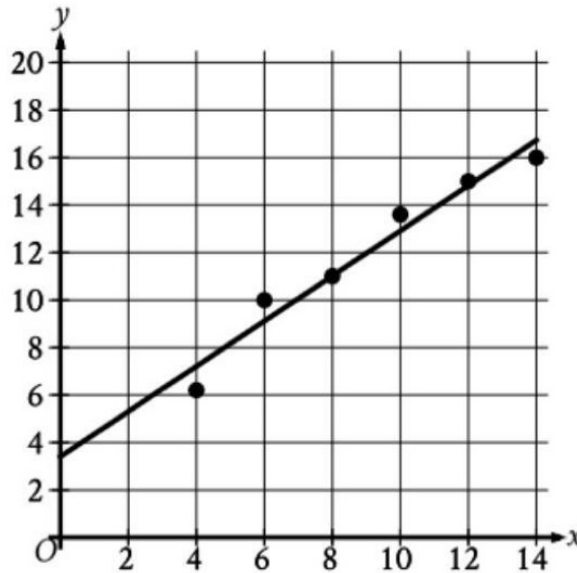
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Question # ID
SAT10 M 1.4 a10ma104

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.

Which of the following equations best represents the line of best fit shown?



(A) $y = x + 3.4$

(B) $y = x - 3.4$

(C) $y = -x + 3.4$

(D) $y = -x - 3.4$

SAT10 M 1.5 a10ma105

Of 300,000 paper clips, 234,000 are size large. What percentage of the paper clips are size large?

(A) 22%

(B) 33%

(C) 66%

(D) 78%

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SAT10 M 1.6 ba79f10f

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $y = 18x + 13$
- B. $y = 18x + 18$
- C. $y = -5x + 13$
- D. $y = -5x + 18$

SAT10 M 1.7 a10ma107

The function f is defined by $f(x) = x^2 + x + 71$. What is the value of $f(2)$?

SAT10 M 1.8 f5c3e3b8

Which expression is equivalent to $(m^4q^4z^{-1})(mq^5z^3)$, where m , q , and z are positive?

- A. $m^4q^{20}z^{-3}$
- B. $m^5q^9z^2$
- C. $m^6q^8z^{-1}$
- D. $m^{20}q^{12}z^{-2}$

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SAT10 M 1.9 a10ma109

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

(A) 204

(B) 182

(C) 60

(D) 12

SAT10 M 1.10 3726e079

If $\frac{x}{y} = 4$ and $\frac{24x}{ny} = 4$, what is the value of n ?

SAT10 M 1.11 56dc8045

$$w(t) = 300 - 4t$$

The function w models the volume of liquid, in milliliters, in a container t seconds after it begins draining from a hole at the bottom. According to the model, what is the predicted volume, in milliliters, draining from the container each second?

A. 300

B. 296

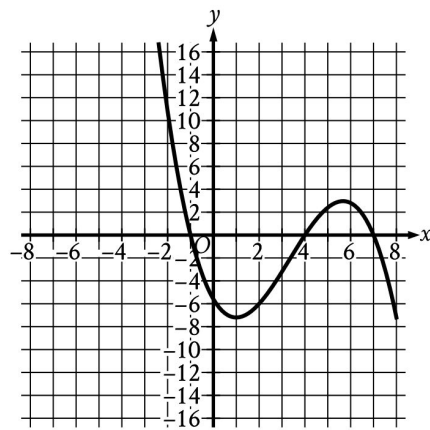
C. 75

D. 4

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Question # ID
SAT10 M 1.12 cc6ccd71



The graph of $y = f(x)$ is shown, where the function f is defined by $f(x) = ax^3 + bx^2 + cx + d$ and a , b , c , and d are constants. For how many values of x does $f(x) = 0$?

- A. One
- B. Two
- C. Three
- D. Four

SAT10 M 1.13 a10ma113

$$f(x) = 3,000(0.75)^x$$

A conservation scientist implemented a program to reduce the population of a certain species in an area. The given function estimates this species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 3,000 in this context?

- (A) The estimated percent decrease in the population for this species and area every 8 years after 2008
- (B) The estimated percent decrease in the population for this species and area each year after 2008
- (C) The estimated population for this species and area 8 years after 2008
- (D) The estimated initial population for this species and area in 2008

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Question # **ID**
SAT10 M 1.14 ee857afb

$$y = x^2 - 14x + 22$$

The given equation relates the variables x and y . For what value of x does the value of y reach its minimum?

SAT10 M 1.15 e5c57163

Square A has side lengths that are 166 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

SAT10 M 1.16 a10ma116

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is (a, b) . What is the value of b ?

SAT10 M 1.17 8490cc45

The function f is defined by $f(x) = (-8)(2)^x + 22$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. $(0, 14)$
- B. $(0, 2)$
- C. $(0, 22)$
- D. $(0, -8)$

SAT10 M 1.18 a8ae0d22

Two variables, x and y , are related such that for each increase of 1 in the value of x , the value of y increases by a factor of 4. When $x = 0$, $y = 200$. Which equation represents this relationship?

- A. $y = 4(x)^{200}$
- B. $y = 4(200)^x$
- C. $y = 200(x)^4$
- D. $y = 200(4)^x$

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Question # ID
SAT10 M 1.19 a10ma119

What is the value of $\sin 42\pi$?

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$

(D) 1

SAT10 M 1.20 14360f84

$$\begin{aligned} 4x - 9y &= 9y + 5 \\ hy &= 2 + 4x \end{aligned}$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

- A. -9
- B. 0
- C. 9
- D. 18

SAT10 M 1.21 ba0edc30

$$x^2 - 2x - 9 = 0$$

One solution to the given equation can be written as $1 + \sqrt{k}$, where k is a constant. What is the value of k ?

- A. 8
- B. 10
- C. 20
- D. 40

SAT10 M 1.22 104bff62

$$\frac{x^2}{\sqrt{x^2 - c^2}} = \frac{c^2}{\sqrt{x^2 - c^2}} + 39$$

In the given equation, c is a positive constant. Which of the following is one of the solutions to the given equation?

- A. $-c$
- B. $-c^2 - 39^2$
- C. $-\sqrt{39^2 - c^2}$
- D. $-\sqrt{c^2 + 39^2}$